MCP-referenced timing calibration of CMS BTL prototype bars

CMS

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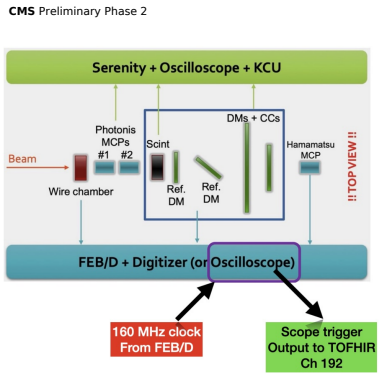
Discussion draft; not approved for public release.

Abstract

A timing-calibration procedure is studied using CMS barrel timing layer (BTL) prototype scintillator bars in the September 2025 beam test. A microchannel-plate photomultiplier (MCP-PMT), digitized by an oscilloscope, provides an external arrival-time reference. A trigger link identifies coincident events in the BTL readout and the oscilloscope. Segment-dependent clock alignment and a locally weighted smoothing correction are used to map the reference into the BTL time coordinate.

The dependence of the measured arrival time on pulse charge is described by quadratic functions of inverse charge for four silicon-photomultiplier channels on two reference bars. These functions provide an empirical time-walk calibration over the measured charge range. Clock-corrected profiles provide the reference for the companion dual-threshold study; archived fits are retained to document the change in calibration procedure. Timing-difference distributions illustrate the validation strategy and the need to distinguish intrinsic reference-detector resolution from residual synchronization effects. This draft documents the available results and identifies the remaining selection, uncertainty and reproducibility checks before publication.

Beam-test setup and the reference signal



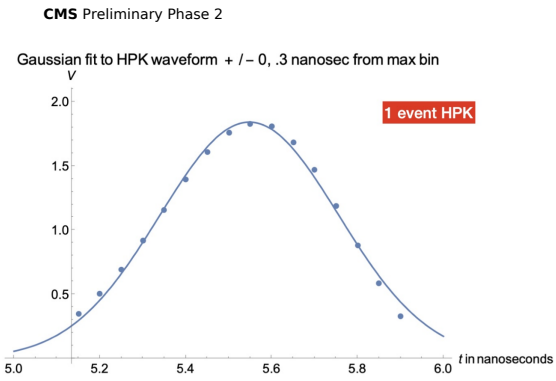
The September 2025 beam test used scintillator bars read out at both ends by silicon photomultipliers (SiPMs). A downstream MCP-PMT provides a fast reference for a restricted region of the bars.

The BTL front-end readout and oscilloscope use different time coordinates. The scope trigger reaches TOFHIR channel 192; a separate 160 MHz clock link can also be recorded by the scope.

The highlighted lower readout path is used for the reference-bar calibration.

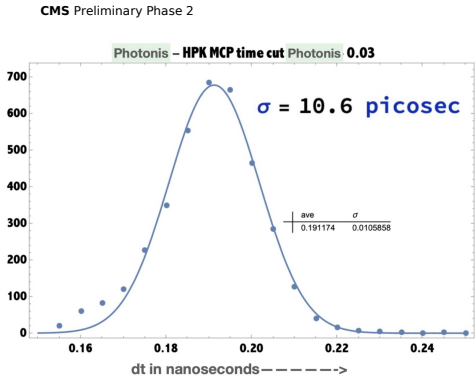
Legacy setup illustration. Detector/readout context: Refs. [1, 2]; run-specific setup from this study.

Reconstructing the MCP pulse time



An example Hamamatsu MCP waveform is shown with its fitted pulse template. The fit estimates the pulse centroid rather than using the threshold-crossing time alone, reducing sensitivity to pulse amplitude.
The scope-trigger time identifies the event; the fitted peak supplies its relative reference time. Their difference is measured within the same oscilloscope waveform.

Intrinsic reference timing is a separate measurement

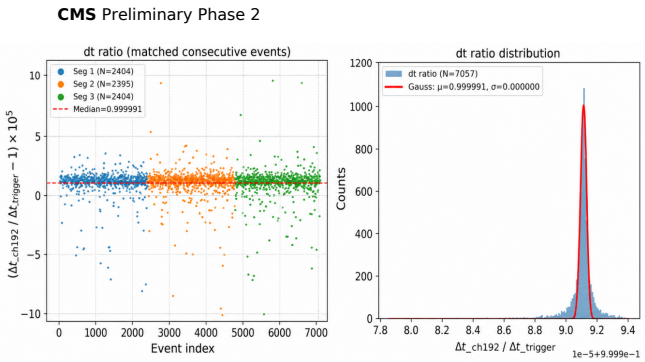


The distribution compares the fitted times of two MCP-PMTs recorded by the oscilloscope. The measured width is 10.6 ps.

Assuming independent errors and a 7.5 ps PHOTONIS contribution gives $\sigma_{\mathrm{HPK}} \simeq \sqrt{10.6^2 - 7.5^2} = 7.5$ ps.

This scope-only estimate does not include synchronization to the BTL readout.

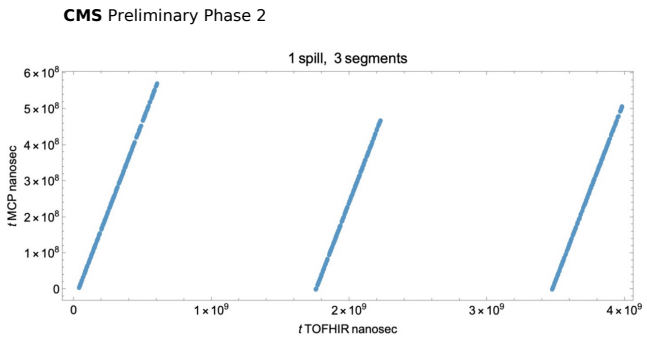
Matching the same particles between readouts



Consecutive trigger intervals measured by channel 192 are compared with those recorded by the scope. Correlated intervals identify candidate matches despite the independent clocks.
The interval-ratio distribution tests matching consistency. It is not a measurement of the picosecond resolution of the reconstructed particle time.
Only uniquely matched events enter the reference calibration.

Implementation (repository-relative):
`FEBD_Sync/simple_sync.py`
`dedup_triggers`: duplicate removal.
`match_cumulative_time`: matching and tolerance.

Clock-rate alignment within acquisition segments

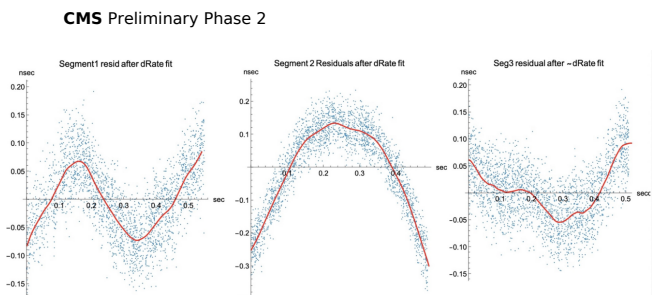


Matched scope-trigger times are compared with TOFHIR trigger times. A nearly linear relation describes the overall clock-rate difference within an acquisition segment.

An offset and a slope are needed: a slope close to one is not sufficient over a long time interval. Clock resets require separate segment coordinates.

Legacy clock-relation plot. This plot illustrates synchronization, not charge-dependent time walk.

Correcting smooth residual clock drift



After the linear clock-rate correction, the time residual still changes within each of three segments. Points are matched events; red curves show the smooth trend estimated by LOWESS, a locally weighted regression [3].
The fitted trend is added to the linear prediction to reconstruct the expected channel-192 trigger time. This expected time is subtracted once from the SiPM time in the reference difference.

Legacy residuals in ns versus segment time in seconds. These are pre-LOWESS residuals, not final timing resolutions. Implementation: `TimeCalibration/ch192_vs_trigger_lowess.py`, lines 657–668 and 816.

A single, explicit time convention

Let $g(u)$ map a scope-trigger timestamp u into the TOFHIR time coordinate, including the linear clock relation and the fitted smooth drift. Define the within-waveform difference

$$\delta_{\mathrm{MCP}} = t^{\mathrm{scope}}_{\mathrm{peak}} - t^{\mathrm{scope}}_{\mathrm{trigger}}.$$

The reference difference implemented in the calibration is

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t^{\mathrm{scope}}_{\mathrm{trigger}}) - \delta_{\mathrm{MCP}},$$

up to a fixed cable/geometry offset and the negligible-rate-conversion approximation for the short within-waveform interval.

The charge-dependent component of Δt_i is the time walk. An arbitrary constant does not change a shape calibration.

Implementation: `TimeCalibration/ch192_vs_trigger_lowess.py`, lines 816 and 1102. Both per-file and combined-run calculations subtract (`mcp_peak - mcp_trig`).

Reference bars and calibration selection

Upstream bar	SiPM channels	Role
7	133, 154	First reference pair
8	137, 150	Second reference pair

- Select coincident SiPM hits and a unique MCP-associated event, using channel 192 as the TOFHIR indicator.
- Require a successfully reconstructed MCP pulse and consistent peak-to-trigger timing. This is a waveform-quality cut, not a particle transit-time measurement inside the MCP.
- Use Q, the TOFHIR ADC charge proxy. It is not a deposited energy in MeV.
- Fit the profile of Δt_i versus $1/Q$ over the retained charge range.

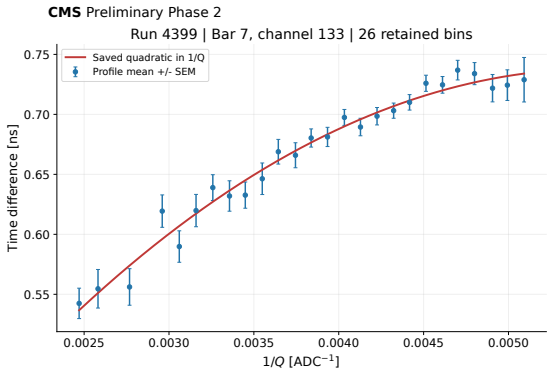
Implementation (repository-relative):

`TimeCalibration/ch192_vs_trigger_lowess.py`: coincidence and walk-sample selection.

`TimeCalibration/ch192_vs_trigger.py`: `build_mcp_map`, optional MCP cuts.

`TimeCalibration/bar_processing.py`: peak-to-trigger timing selector.

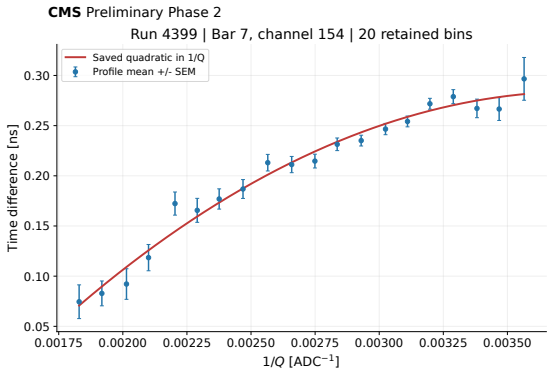
Archived reference: bar 7, channel 133



Run 4399 events are grouped in inverse-charge bins. Each point is the mean time difference; the vertical bar is its saved standard error of the mean (SEM). The curve is the saved binned quadratic $W(Q) = a + b/Q + c/Q^2$. There are 26 retained profile points. The time coordinate includes the historical constant display offset. These archived means use measured channel-192 timing. An unweighted fit to the means reproduces the saved coefficients; SEM bars are not fit weights.

Redrawn from archived profile and coefficient CSVs. No refit. Saved display convention includes a +26 ns shift.

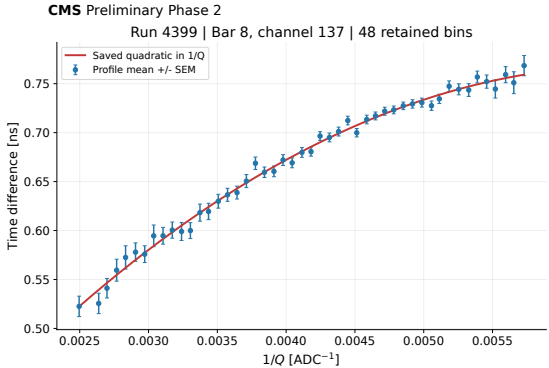
Archived reference: bar 7, channel 154



The opposite end of bar 7 is calibrated independently. Points show the mean time difference and SEM in 20 retained inverse-charge bins.
The red curve is the saved quadratic in $1/Q$, not a quadratic in Q.
Channel-dependent coefficients allow for different pulse responses at the two bar ends. The archived fit is unweighted and uses measured channel-192 timing. The retained bin counts are channel dependent.

Run 4399 legacy calibration, redrawn without refitting. Same +26 ns display convention as channel 133.

Archived reference: bar 8, channel 137

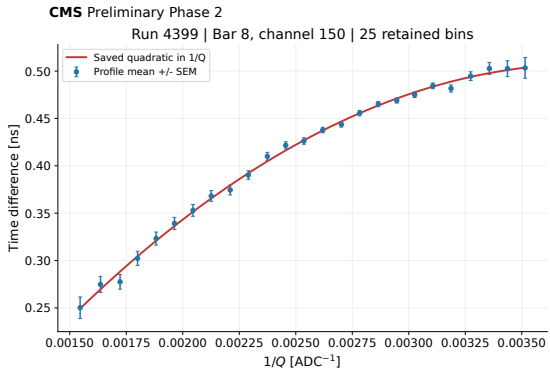


The channel-137 time difference increases as inverse charge increases: smaller signals cross a fixed threshold later.

Points are profile means with SEM; the red line is the saved binned quadratic. There are 48 retained points. The correction is applicable over the measured charge support, not automatically outside it. This unweighted, measured-channel-192 result is retained for comparison with the clock-corrected reference shown next.

Run 4399 legacy calibration. Coefficients and numerical points are included with the draft.

Archived reference: bar 8, channel 150



The opposite end of bar 8 has its own charge-dependent delay. The 25 points show the mean timing difference and SEM; the curve is the saved binned quadratic.

The four curves demonstrate the calibration procedure. Different numbers of retained points are preserved rather than imposing an undocumented common binning. This is the archived, unweighted fit using measured channel-192 timing.

Run 4399 legacy calibration. Coefficient units are ns, ns ADC and ns ADC^2 for a, b and c.

Clock-corrected reference shared with Note 2

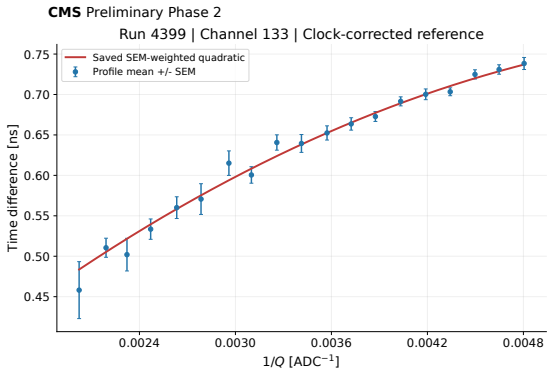
The following results use the expected trigger time $g(u)$ reconstructed from the segment-wise linear clock relation and LOWESS drift:

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t_{\mathrm{trigger}}^{\mathrm{scope}}) - (t_{\mathrm{peak}}^{\mathrm{scope}} - t_{\mathrm{trigger}}^{\mathrm{scope}}).$$

- Run 4399: exactly five TOFHIR hits, comprising the target upstream pair, downstream channels 234 and 243, and channel 192, with a unique MCP match.
- Require finite, positive charge and slope-related quality quantities; retain Q at or above the left half-maximum boundary of the spectrum peak.
- Form 20 uniform $1/Q$ bins; retain times within $5(1.4826\,\mathrm{MAD})$ of each bin's median, with at least five entries. Bars show the SEM after clipping.
- Fit $W(Q) = a + b/Q + c/Q^2$ with an objective weighted by $1/\mathrm{SEM}^2$. The next four figures use the exact saved reference coefficients used in Note 2.

Source snapshots: `2026-10-22_clock_quality/inputs`. The displayed $+26$ ns constant changes neither the shape nor the saved coefficients.

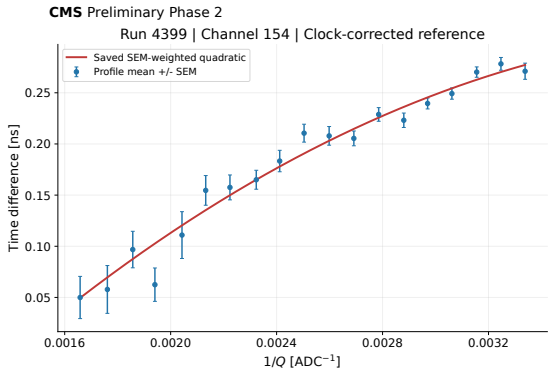
Clock-corrected reference: bar 7, channel 133



The time difference now uses the reconstructed trigger time rather than the measured channel-192 timestamp, suppressing the fitted clock drift.
Points show mean time differences with SEM in 19 retained inverse-charge bins.
The red curve is the saved SEM-weighted quadratic used in Note 2.
The curve is shown only within retained-bin support and the outer profile means. Missing bins are not extrapolated across.

Run 4399; frozen Note2 input, redrawn without refitting. A constant +26 ns shift is applied only for display.

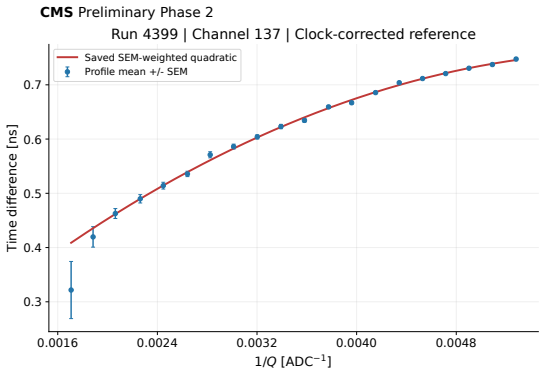
Clock-corrected reference: bar 7, channel 154



The opposite end of bar 7 is calibrated with the same clock-reconstruction and profile procedure. Its independent coefficients describe its own charge-dependent timing response. The 19 points are retained-bin means; vertical bars show SEM after timing clipping. The red curve is the saved uncertainty-weighted quadratic in $1/Q$.

Exact reference snapshot used in Note2. Coefficients and profile points are included with this note; the +26 ns display shift is separate.

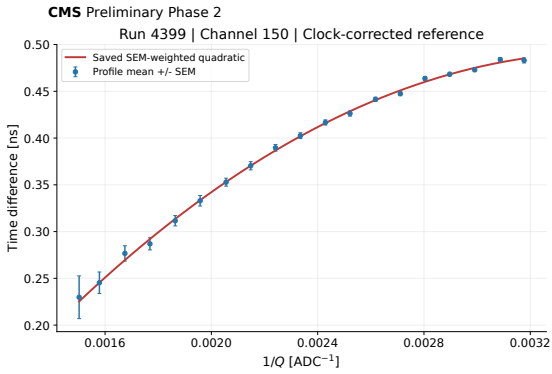
Clock-corrected reference: bar 8, channel 137



There are 20 retained inverse-charge bins for this channel. Smaller charge gives a larger timing delay, as expected for fixed-threshold readout of pulses with different amplitudes.
Points and bars are the mean and SEM after clipping. The saved quadratic is the MCP benchmark for the channel-137 inverse-slope comparison in Note 2.

Run 4399; expected-trigger timing; refitted after clock-quality selection. The +26 ns shift affects only the vertical display origin.

Clock-corrected reference: bar 8, channel 150



The channel-150 reference uses 19 retained profile bins and the same clock-corrected time definition as the other three channels.
Bars represent SEM, not the spread of individual event times. The red line is the saved SEM-weighted inverse-charge quadratic. These four results connect the MCP calibration in this note to the AWC shape comparison in Note 2.

Run 4399; coefficients are preserved in their original ns convention. Only the figure receives the constant +26 ns display shift.

Archived and clock-corrected shapes are close

Compare both saved functions on jointly populated charge support, after removing a constant offset only. No multiplicative scaling is fitted.

Channel	Shape RMS difference [ps]	Maximum absolute difference [ps]
133	3.20	10.63
154	4.29	9.68
137	1.33	3.22
150	4.91	10.47

The current reference includes finite-timestamp reset handling and a clock-only quality requirement. The archived and current versions also differ in trigger reconstruction, selection, clipping and fit weights.

These are differences between calibration functions, not timing resolutions or a statistical-equivalence test. Note 2 uses the clock-corrected version.

Clock quality before timing validation

Consecutive finite scope timestamps define segment boundaries. Missing timestamps cannot conceal a reset; nonfinite triggers have no valid mapping.

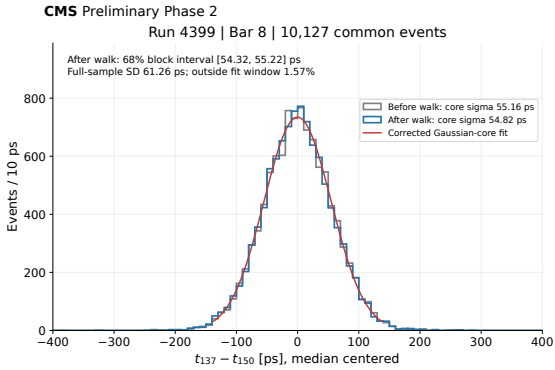
For each topology, file and segment, fit channel 192 against the scope trigger. Reject the whole group if the line-residual scale $1.4826\,\mathrm{MAD}$ exceeds 1 ns, before LOWESS. This rule uses clocks only, not SiPM arrival times.

Six finite-trigger bar-8 groups fail. Changing the threshold from 1 ns to 1 μs retains the same groups. Suspected one-event association offsets are excluded, not repaired by shifting MCP rows.

Refit the channel walk curves after this selection. For the following timing comparison require both ends to pass quality, peak-left charge cuts and their own populated reference support within the outer profile means. No additional selection is made on either timing-difference observable.

Implementation: `Q_Walk/qwalk_t0_data.py`; new references: `2026-10-22_clock_quality`.

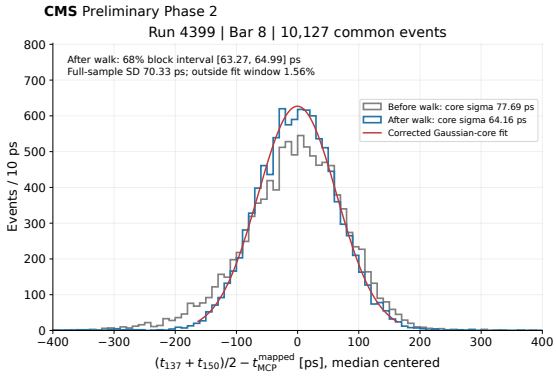
Timing validation using the two bar ends



The same 10127 bar-8 events are shown before and after subtracting the two channel-specific walk functions. The common clock mapping is fixed. The Gaussian-core width changes from 55.16 to 54.82 ps. The corrected full-sample SD is 61.26 ps; 1.57% lie outside the core window.
The 16th–84th percentile interval is [54.32, 55.22] ps from 400 file/segment-block bootstrap replicas, with calibrations held fixed. Ref. [4] gives the general resampling principle.

Truncated-normal likelihood within median $\pm 2.5(1.4826\,\mathrm{MAD})$. Each histogram is median-centered for display only; the SD includes all events.

Timing validation against the MCP reference



The bar-average time minus mapped MCP time uses the identical event sample and clock reconstruction before and after channel-wise walk correction. The core width changes from 77.69 to 64.16 ps. The corrected full-sample SD is 70.33 ps; 1.56% lie outside the fit window.

The conditional block-bootstrap interval is [63.27, 64.99] ps. This observable includes detector, MCP and residual clock contributions.

Same core-fit definition as the end difference. The display range does not restrict the SD. No segment-by-segment timing recentering is applied.

What these timing widths establish

The reproduced distributions establish conditional before/after performance, not intrinsic bar resolution or a separate synchronization-jitter measurement. At fixed position, for end timing errors e_L, e_R,

$$\mathrm{Var}\Big(\frac{e_L + e_R}{2}\Big) = \frac{\mathrm{Var}(e_L - e_R)}{4} + \mathrm{Cov}(e_L, e_R).$$

Halving an end-difference width requires controlled position effects and negligible end-error covariance. Shared MCP and clock errors also prevent inferring detector covariance directly from the measured reference residuals.

The two separately fitted core widths are not variances of a common model. Their quadrature difference is not a justified synchronization-jitter estimate. This version reports same-event timing validation with conditional statistical intervals. Intrinsic bar resolution and synchronization-jitter extraction are outside its scope. Position, covariance, segment-offset and calibration systematics are not included in the quoted intervals.

Reference-calibration result and scope

- A trigger link permits common-event calibration across independent BTL and oscilloscope time coordinates.
- A linear clock relation plus a smooth residual correction defines the reference-time mapping.
- Four clock-corrected inverse-charge profiles and quadratic functions provide the reference used in Note 2. The archived fits remain as a comparison; numerical points, SEM and both coefficient sets accompany this draft.
- Intrinsic MCP timing, reference-bar timing and cross-system synchronization are different performance quantities.
- Charge range and operating conditions limit transfer to other samples; a constant time offset is not a walk-shape difference.

For the same selected events, walk correction reduces the bar–MCP core width from 77.69 to 64.16 ps. This is a conditional timing-validation result, not an intrinsic-resolution measurement or a complete uncertainty budget.

References

[1] CMS Collaboration, *A MIP Timing Detector for the CMS Phase-2 Upgrade*, CERN-LHCC-2019-003, CMS-TDR-020 (2019). CERN document record.

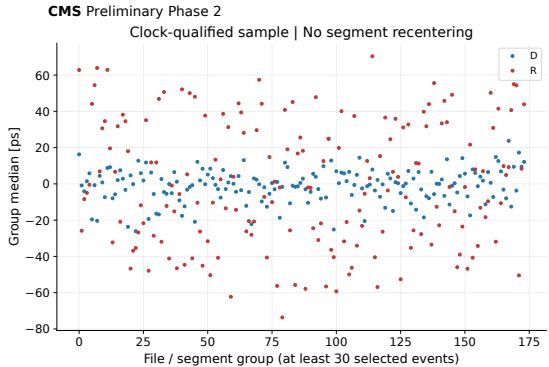
[2] E. Albuquerque et al., *TOFHIR2: The readout ASIC of the CMS Barrel MIP Timing Detector* (2024). arXiv:2404.01208.

[3] W. S. Cleveland, *Robust Locally Weighted Regression and Smoothing Scatterplots*, J. Am. Stat. Assoc. **74** (1979) 829–836. doi:10.1080/01621459.1979.10481038.

[4] B. Efron, *Bootstrap Methods: Another Look at the Jackknife*, Ann. Statist. **7** (1979) 1–26. doi:10.1214/aos/1176344552.

These sources provide detector/readout context and methodological background. The run-specific selections, implementations and numerical results are those of this study, not measurements or prescriptions taken from these references.

Backup: residual segment offsets



Points show median end differences (D) and bar-minus-MCP differences (R) for groups containing at least 30 selected events.

Clock-quality selection removes gross reconstruction failures, but smaller group offsets remain. They contribute to the aggregate timing distribution.

The main timing figures do not subtract these medians. Their effect must be understood before extracting a synchronization contribution.

Group medians are descriptive point estimates, not independent resolution measurements. Identifiers and event counts accompany the numerical data.